

detail will ensure your solutions do not remain in lab but are good enough for use by the masses and would work for years to come. You may work with other ventures as you partners or sub-contractors, if you do not have adequate in-house support for installation and after-sales.

Security

Just like vaccine has become an important part of our lives during current pandemic, security is like the vaccine that must go in each and every IoT system. As the idea of networking appliances and other objects is relatively new, security is not always considered top priority during a product's design phase. Because the IoT solution providers want to get their products to market quickly, they often forget to take the necessary steps to build security from the start.

After deployment of an IoT system, if there are any security attacks—which is bound to happen sooner or later—it would be extremely painful for the organisation using it as well as the startup. A common IoT device in a highly integrated system can be easily used to infiltrate and attack the larger network, so it is critical to ensure safety of the network and IoT devices connected to it. There are various techniques, strategies, and protocols to ensure that the security is up to the mark.

Personality

After ensuring the wellness of a human body, it is important to develop a personality, which in terms of IoT refers to being a significant provider in one of the domains. Pick up a domain that you can work on the best, build solutions, earn money, and only then start expanding to the other domains. It is important not to be a jack of all trades. For startups, especially, focusing on a par-



ticular domain, getting the right people from the domain, and start deploying the solution is the key.

There are various domains in which one can deploy solutions, but there are some technology trends that should not be ignored such as NB-IoT and 5G, which are going to create a big revolution in the IoT sector. Following are some other technology trends that should not be ignored:

1. Edge and Distributed Computing
2. Internet of Behaviour
3. E2E Security
4. Analytics, AI (Artificial Intelligence), and ML (Machine Learning)
5. Healthcare Sensors
6. Electric Vehicles and Autonomous Cars
7. Agriculture
8. Digital Twins
9. Blockchain

The coming decade

In the coming decade, we can expect a shift in the way IoT is implemented in our lives. IoT will find its place everywhere from consumer based IoT products to industrial adaptation. The success of consumer based IoT devices will lead to its pilot use across different industries.

To meet the requirements of the

coming decade, startups and ventures must gear up and work on all aspects of an IoT ecosystem to make sure that the solution delivered is effective in all ways. Startups should have a team that also has, besides software developers, network specialists, hardware developers, and domain experts.

It is important to choose the most optimum communication technology that can scale, is available across the globe, and needs no infrastructure deployment cost. Of course, longevity is also an aspect to be kept in mind. The implicit IoT requirements such as security, device management/FOTA, power management, and hardware watchdogs should not be ignored.

As mentioned earlier, do not ignore installation and after-sales service. Apart from this, make sure to choose open source technologies over proprietary ones for development, as open source provides your scalability; you can expand what you currently do quickly. And, finally, make sure to democratize IoT with ease and economy of adoption. **EFY**

The article is based on a presentation by Anand Bhandari, Head of NB-IoT, Smart Utilities and Smart Assets at JioThings. It has been prepared by Darshil Patel, a Technology Journalist at EFY until recently.